



Turn YouTube lessons into real mastery.

Paste a public YouTube lesson, build an evidence-grounded quiz inside your own browser, and learn through immediate corrective feedback.

LIVE PRODUCT — OPEN THIS FIRST

<https://clipquest.ccwu.cc>

Source code: <https://github.com/404-Waifu-Not-Found/HHCC2026>

HHCC 2026

ITERATION TRACK

EDUCATION THEME

Field	Entry
Project name	ClipQuest
Team	404-Waifu-Not-Found · @404-Waifu-Not-Found/cos
Members	UnoxyRich · Layla · Justin-Yonardo · JimmyfaQwQ
Track	Iteration Track
Theme	Education — accessible, engaging learning experiences with AI feedback
Live product	https://clipquest.ccwu.cc
Repository	https://github.com/404-Waifu-Not-Found/HHCC2026
Document	Project description & iteration report — 22 August 2026

Pre-event baseline — please read before judging

One week before HHCC 2026, this project existed **only as a written README** describing the product idea. There was no implemented product flow, no frontend, no quiz engine, no persistence layer, no browser extension, no native client, and no deployment. **Every line of product code in this repository was written during the 36-hour challenge.** The Git history is the proof: all 490 commits carry timestamps inside the official hacking period of 21–22 August 2026.

490

COMMITTS IN 36H

90,635

LINES OF CODE

4

RUNTIME
SURFACES

26

DB MIGRATIONS

159

AUTOMATED
CHECKS

Contents

1. Executive summary
2. Pre-event foundation and in-event declaration
3. Development background — the problem we set out to solve
4. Core concept — how ClipQuest works
5. Functional modules
6. Learning-science mapping
7. Product walkthrough with screenshots
8. System architecture
9. Technology stack
10. Reliability, correctness, and recovery engineering
11. Privacy and security boundary
12. Original visual system and brand
13. Iteration log — what changed during the 36 hours
14. Pre-event versus post-event comparison
15. Verification, QA, and honest release status
16. Team and responsibilities
17. AI usage disclosure
18. Social impact, sustainability, and business model
19. Current progress and future plans
20. Appendix — repository structure and links

1. Executive summary

ClipQuest turns any public YouTube lesson into an evidence-grounded mastery quest. A learner pastes a YouTube link, chooses the question types they want — multiple-choice, true/false, short answer — confirms the lesson, and immediately starts answering questions that were generated from the video's own captions. Every answer receives immediate corrective feedback; every completed lesson becomes durable progress with mastery ranks, scheduled review, and a personal Library.

The architecturally unusual decision in ClipQuest is *where the AI runs*. Generation does not happen on our server. A Chrome Manifest V3 extension — the ClipQuest Local AI bridge — acquires the video's public captions inside the learner's own browser, normalizes them to plain text, and calls DeepSeek directly using the **learner's own API key**. Our Cloudflare Worker never sees the key, the captions, the prompt, or the raw model output. It only ever receives questions that have already been validated locally, and it stores them in strict ordinal order.

That boundary is not just a privacy feature. It is what makes the product economically sustainable — our per-learner inference cost is exactly zero — and it is what forced the hardest engineering in the project: if the model runs on the far side of an untrusted browser, the server has to be able to reject anything malformed without ever falling back to invented content, and the client has to be able to repair a failed question without destroying the learner's progress.

ClipQuest is live at <https://clipquest.ccwu.cc>, deployed on Cloudflare Workers with D1, KV, and R2, and reachable throughout judging.

2. Pre-event foundation and in-event declaration

2.1 What existed before the event

Under the Iteration Track rules, we submitted our pre-event material one week before HHCC 2026. That material was a **short written README describing the product idea** — the concept of converting YouTube lessons into quizzes, and a sketch of the intended learner flow.

What did *not* exist before the event:

- No implemented product flow of any kind.
- No frontend — no React application, no routes, no components, no styling system.
- No quiz engine, no prompt design, no parser, no validator, no retry logic.
- No persistence layer — no database, no schema, no migrations, no authentication.
- No browser extension and no caption-acquisition code.
- No native iOS or Android path.
- No deployment, no domain, no CI, and no test suite.
- No brand, no logo, no icon set, and no visual language.

Declaration

All code in this repository was written during the official 36-hour hacking period. We did not bring a partially built codebase into the event. There is no hidden, commented-out, or pre-staged functional code. The complete Git history is public and every commit is timestamped inside the event window (21–22 August 2026), so this claim is directly verifiable by the organizing committee with `git log --format='%ad %s' --date=iso`.

2.2 Why this makes the iteration measurable

Because the starting point was a written idea rather than a half-finished product, the before-and-after comparison in section 14 is unusually clean. Every row in that table moves from "did not exist" to "shipped and deployed." The iteration log in section 13 records the specific product risk that motivated each stage, so the improvement logic — not just the improvement volume — is visible.

3. Development background — the problem we set out to solve

3.1 The pain point

Educational video is the most abundant free learning resource on earth, and it is also the most passive. A learner can watch a 40-minute lecture, feel that they understood it, and retain very little of it a week later. This gap between *fluency* (it felt easy while watching) and *retention* (can you actually recall it later) is one of the best-documented effects in learning science, and video is unusually good at producing it, because playback never asks the learner to produce anything.

The standard workarounds all fail for different reasons:

- **Taking notes while watching** splits attention and mostly produces transcription rather than retrieval.
- **Making your own flashcards** works, but the effort cost is so high that almost nobody sustains it for casual learning.
- **Existing AI quiz generators** typically send the video title, or a summary, or a scraped description to a language model and ask it to invent questions. The result is questions about a topic rather than questions about *this lesson* — plausible, fluent, and frequently ungrounded in what the instructor actually said.
- **Structured course platforms** have real quizzes, but they are closed gardens. You cannot bring the specific video your professor recommended.

3.2 Our thesis

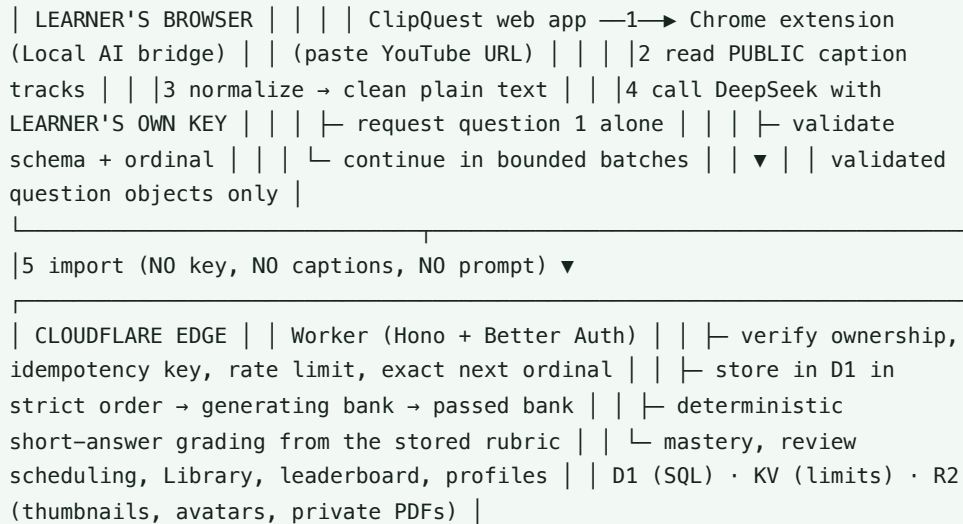
Retrieval practice — being made to produce an answer from memory — is dramatically more effective for durable retention than re-watching or re-reading. So the intervention worth building is not a better summary; it is a **low-friction way to convert any video the learner already chose into retrieval practice**, with the questions grounded in the actual spoken content, and with feedback delivered at the moment the learner is still holding the question in mind.

Three constraints followed from that thesis, and they shaped the entire architecture:

1. **Grounding is non-negotiable.** If the question is not derived from the lesson's real captions, the learner practices the wrong thing. So captions are the only accepted source, and a video without usable captions must fail loudly rather than degrade into invention.
2. **Waiting kills the habit.** If a learner has to wait 60 seconds staring at a spinner before question 1, they will not do this twice. So generation had to become progressive.
3. **Cost must not scale with learners.** A free educational tool that pays per learner per quiz dies at scale. So the inference had to move to the learner's own key and browser.

4. Core concept — how ClipQuest works

ClipQuest is deliberately split so that the untrusted, expensive, privacy-sensitive work happens locally and the trusted, cheap, durable work happens at the edge.



The single most important property of this diagram is the direction of trust. The Worker treats the extension as hostile input. Anything that is malformed, out of order, duplicated, wrongly typed, or answer-inconsistent is rejected — and critically, **rejection never produces a fallback question**. There is no code path in the Worker that can generate quiz content. A learner either gets a real, validated, caption-grounded question, or they get an explicit failure state.

5. Functional modules

Module	What it does
Link-first import	Validates a public YouTube URL, fetches metadata, and renders a lesson preview so the learner confirms they picked the right video before any generation cost is incurred.
Caption acquisition	Reads public caption tracks in-browser, preserves the complete ordered segment set, strips timestamps, joins rolling auto-caption fragments, and emits clean plain text. Never downloads, decodes, or transcribes audio.
Progressive generation	Requests question 1 alone, validates it, and opens the attempt immediately; remaining questions stream in bounded consecutive batches in the background while the learner is already answering.
Quiz engine	Multiple-choice, true/false, and short-answer items in 5-, 10-, or 15-question banks, with a seeded and balanced type plan that avoids long runs of a single format.
Immediate feedback	Every submission is graded and explained at once — correct/incorrect state, the correct answer, and the reasoning — rather than deferred to the end of the quiz.
Deterministic short-answer grading	The Worker grades prose answers from a bounded stored rubric of required ideas and full-credit variants, normalizing pronouns, safe acronyms, and conservative aliases. No model call, so grading is reproducible and the learner's writing never leaves our boundary to a third party.

Completion recap	At the end of an attempt, the learner sees exactly which concepts they missed, with the correct answer and reasoning, forming a targeted review list instead of a bare score.
Library and mastery	Completed lessons persist with colour-coded mastery ranks, progress bars, review state, due reviews, and saved lessons — turning one-off results into a spaced-repetition loop.
Community layer	Server-tracked learning time, a leaderboard, public profiles with avatars and completed-quiz counts, and a GitHub-style yearly activity calendar. Private account details stay hidden.
Operations console	Server-side roles, audited privileged actions, processing overview, and user administration — with generic Better Auth admin endpoints blocked.
Cross-platform clients	Web plus an installable PWA, the Chrome extension bridge, and native iOS and Android paths that share the same local quiz engine and the same privacy boundary.

6. Learning-science mapping

Each learner-facing feature exists because it implements a specific, well-evidenced learning mechanism. This mapping was written before the features, not after them.

Mechanism	ClipQuest feature	Why it is implemented this way
Retrieval practice	Mandatory question answering after (or during) the lesson	Producing an answer from memory strengthens retention far more than re-watching. The learner cannot passively advance.
Immediate corrective feedback	Per-question grading with the correct answer and reasoning	Feedback delivered while the item is still in working memory corrects the misconception rather than reinforcing it.
Generation effect	Short-answer items with deterministic rubric grading	Free recall produces stronger encoding than recognition, so we kept prose answers instead of shipping multiple-choice only.
Desirable difficulty	Balanced mixed-format plans; option order re-randomized per view	Prevents position memorization and pattern-matching, so the learner has to actually retrieve the content.
Error-focused review	Completion recap listing only missed concepts	Directs limited study time at the gap rather than at material already mastered.
Spaced repetition	Library mastery ranks, review state, and due reviews	Distributed practice over time beats massed practice; the Library schedules the return visit.
Consolidation	Per-lesson progress and mastery summary	Gives the learner an accurate model of what they know, countering the fluency illusion that video creates.
Content grounding	Caption-only source text; captionless videos fail closed	Practising a hallucinated fact is worse than not practising. Grounding is a learning requirement, not just a technical one.

7. Product walkthrough with screenshots

All captures below are from the deployed product. The full-resolution set lives in `docs/screenshots/` in the repository.

7.1 Link-first home

The entire product starts from one input. There is no course catalogue to browse and no content to buy — the learner brings the lesson.

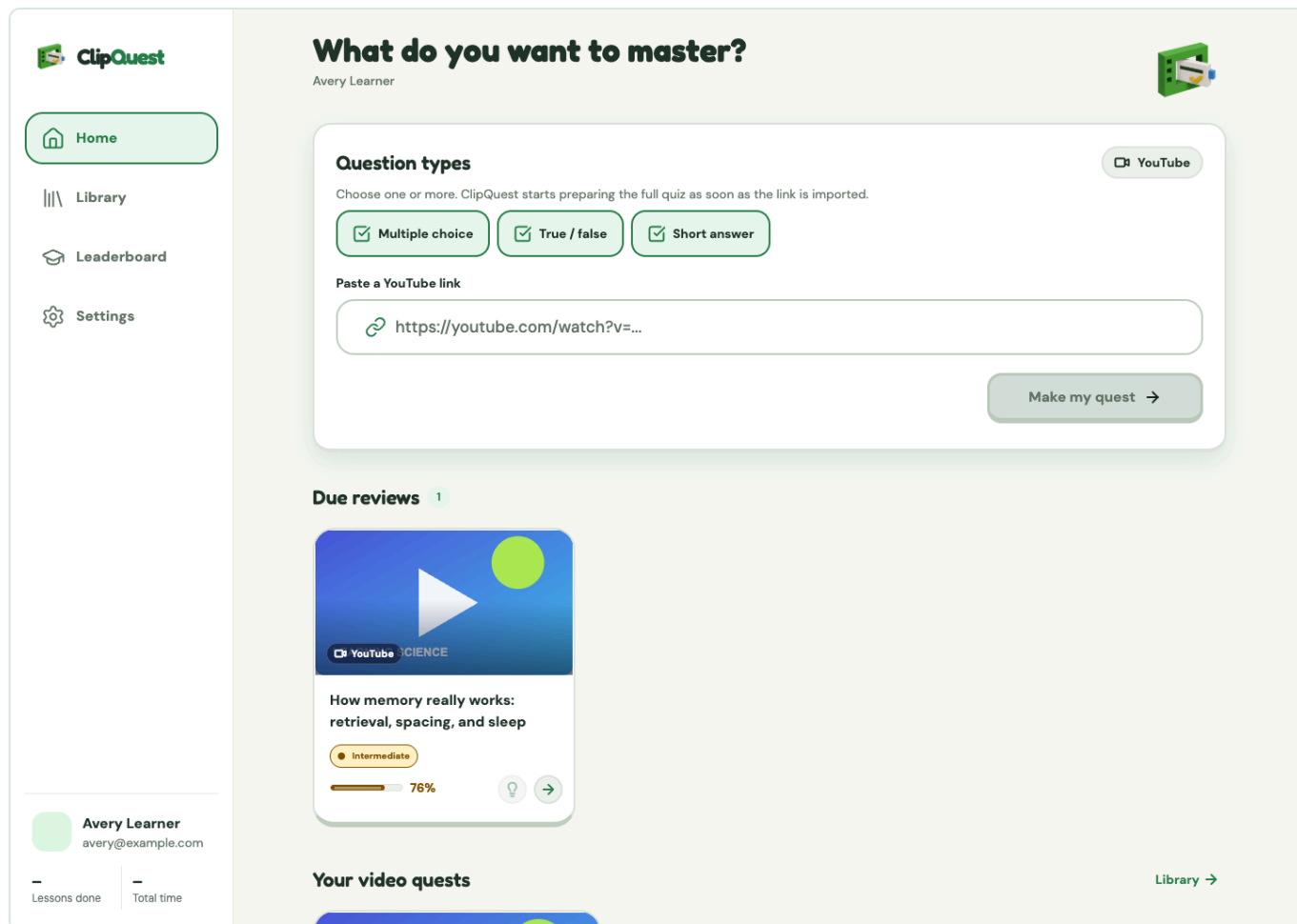


Figure 1 — Link-first import. One field, immediate validation, no catalogue.

7.2 Detection and processing

Once the URL validates, ClipQuest previews the detected lesson so the learner confirms the right video, then shows a time-linear generation clock. That clock is fixed-duration and monotonic by design: internal stages cannot reset it or change its speed, because a progress bar that jumps backwards destroys trust faster than a slow one.

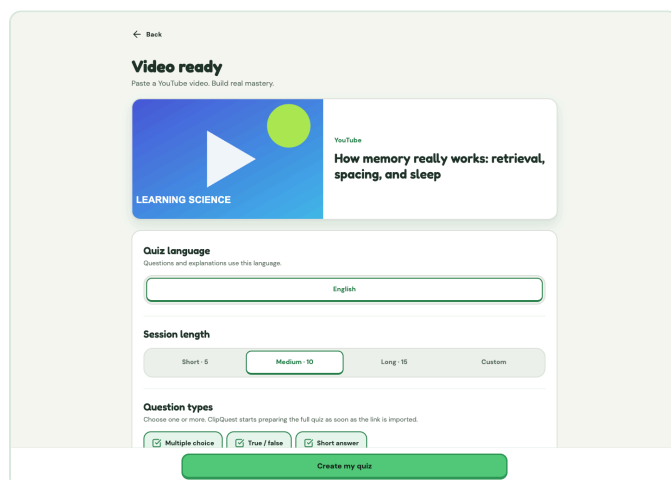


Figure 2 — Detected lesson preview and question-type selection.

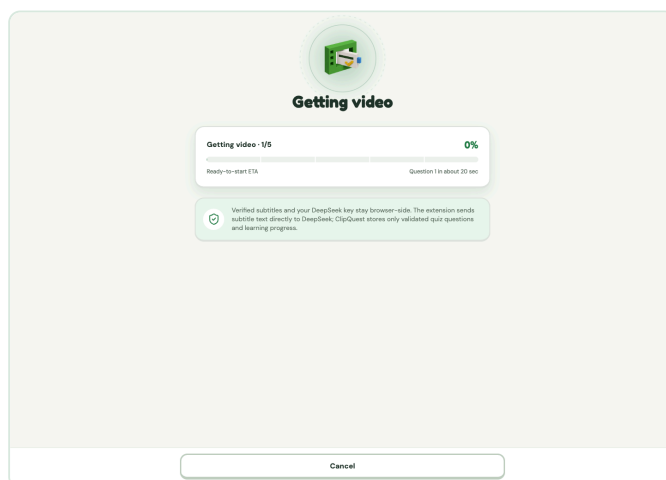


Figure 3 — Time-linear generation clock with explicit stages.

7.3 The generated learning experience

The attempt opens as soon as question 1 is authoritative — the learner starts answering while the rest of the bank is still being generated behind them.

The screenshot displays a user interface for a generated lesson. At the top, there is a close button (X) and a progress indicator showing 1/5. Below this, the instruction "SELECT ONE" is visible. The main question is "Why does retrieval practice strengthen long-term memory?". There are four answer options, each in a rounded rectangular button: A (It makes every video shorter), B (It removes the need for sleep), C (It guarantees perfect recall), and D (It requires the brain to reconstruct the idea). At the bottom center is a "Check answer" button. In the bottom right corner, a small badge indicates "1/10 questions ready".

× 1/5

SELECT ONE

Why does retrieval practice strengthen long-term memory?

A It makes every video shorter

B It removes the need for sleep

C It guarantees perfect recall

D It requires the brain to reconstruct the idea

1/10 questions ready

Check answer

Figure 4 — A generated lesson. Large targets, visible progress, one task in focus.

7.4 Immediate corrective feedback

Correct and incorrect states are semantically distinct, not colour-only, and an incorrect answer always surfaces the correct answer plus the reasoning drawn from the lesson.

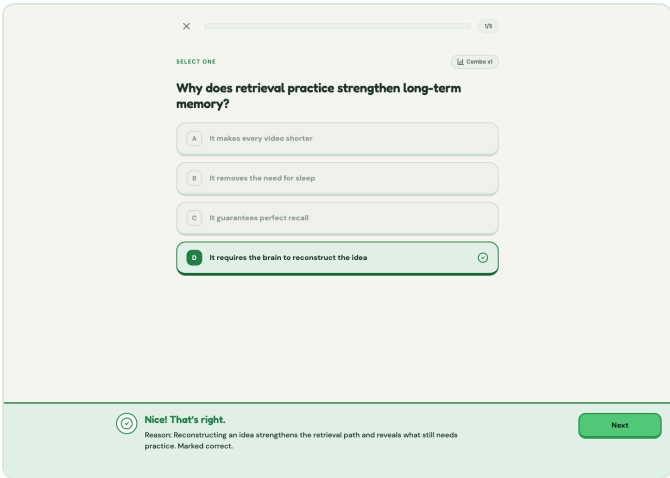


Figure 5 — Correct-answer feedback.

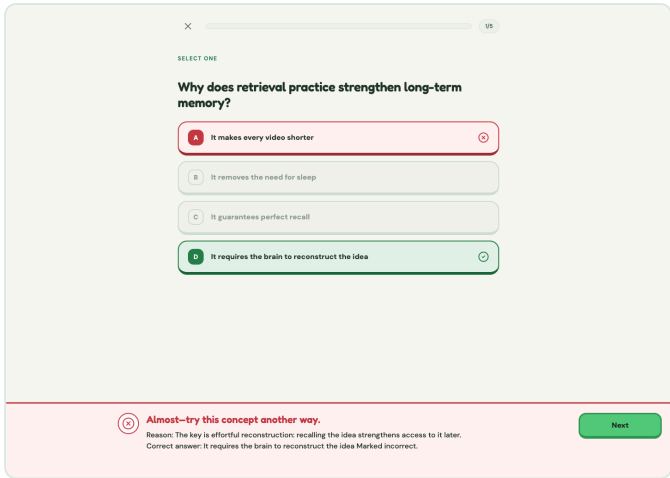


Figure 6 — Incorrect-answer feedback with correction and reasoning.

7.5 Completion recap — what to review

The end of an attempt is not a score screen. It is a study list: the concepts the learner missed, each with the correct answer and the reasoning, ready to be reviewed.

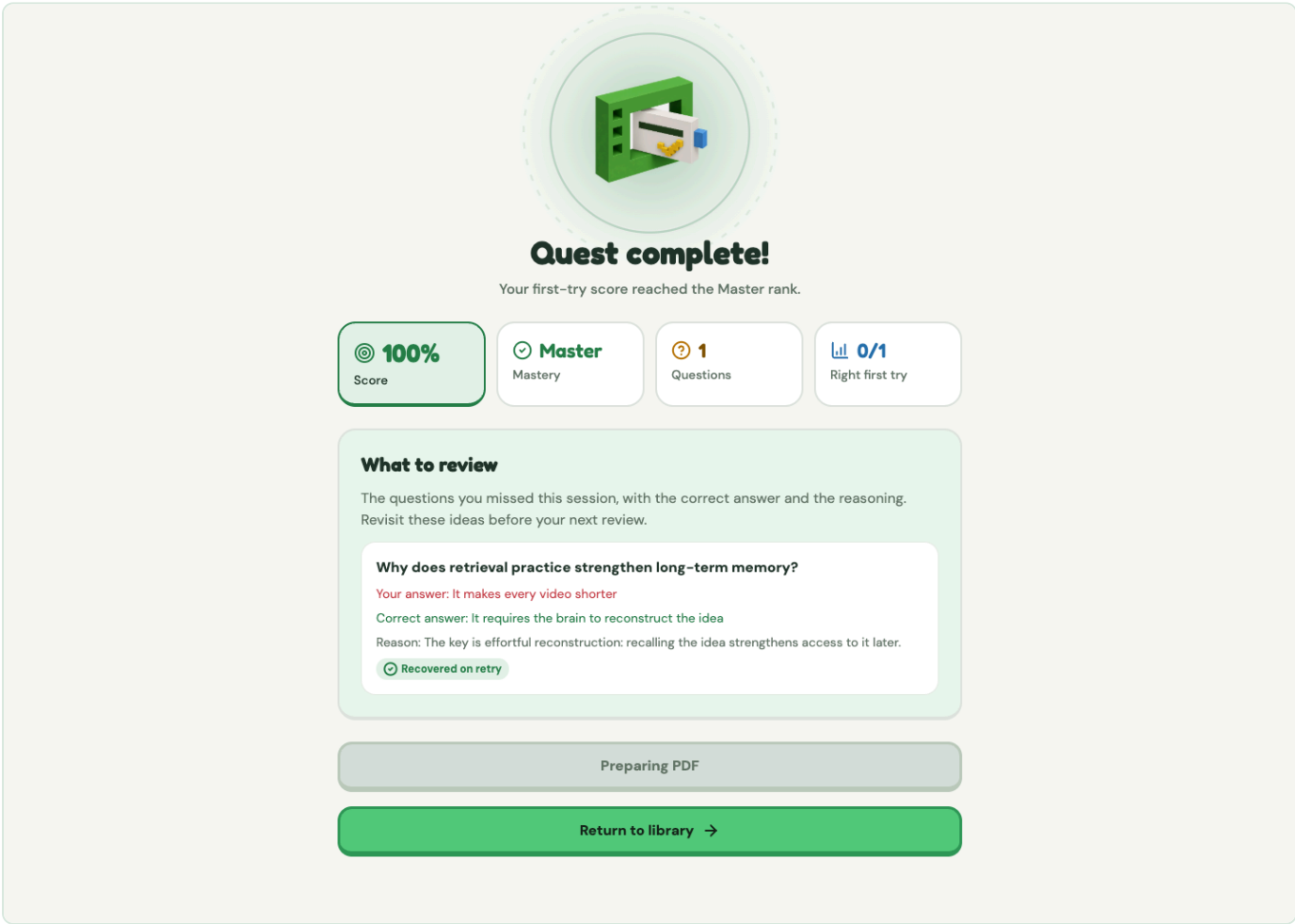


Figure 7 — Missed-concept recap that converts a result into a review plan.

7.6 Library, mastery, and community

Progress persists. The Library holds every completed lesson with colour-coded mastery and due reviews; the leaderboard and public profiles add social accountability with server-tracked learning time and a year-long activity calendar.

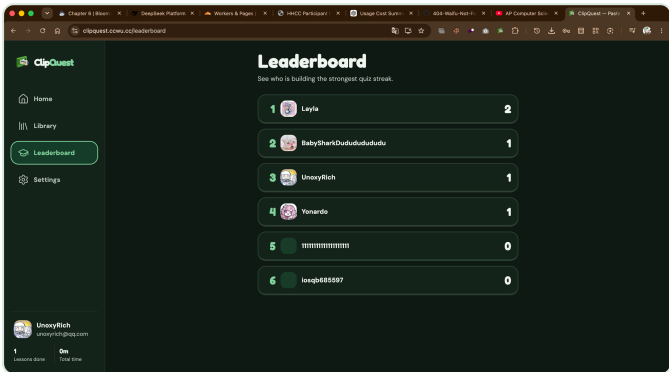


Figure 8 — Library with mastery ranks and progress (live production capture).

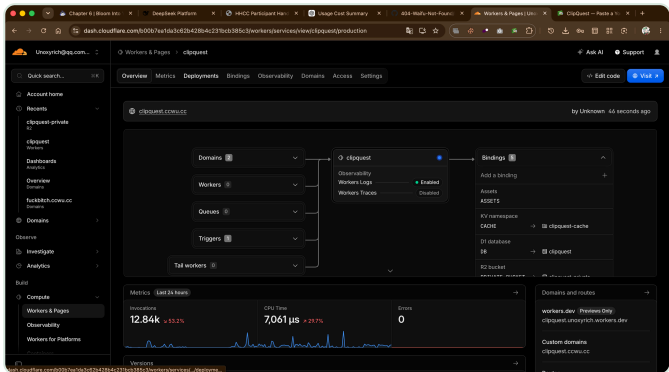


Figure 9 — Leaderboard and public profiles (live production capture).

7.7 Dark theme parity

Light and dark are equally supported first-class themes across learner, authentication, quiz, administration, extension, and PWA surfaces — not an inverted afterthought.

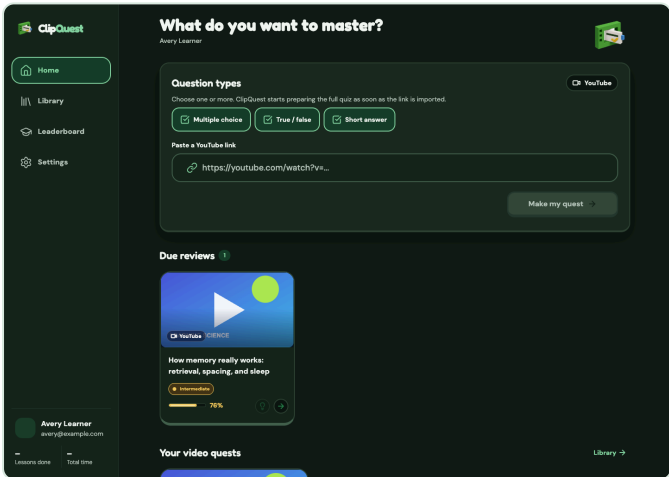


Figure 10 — Dark theme, home.

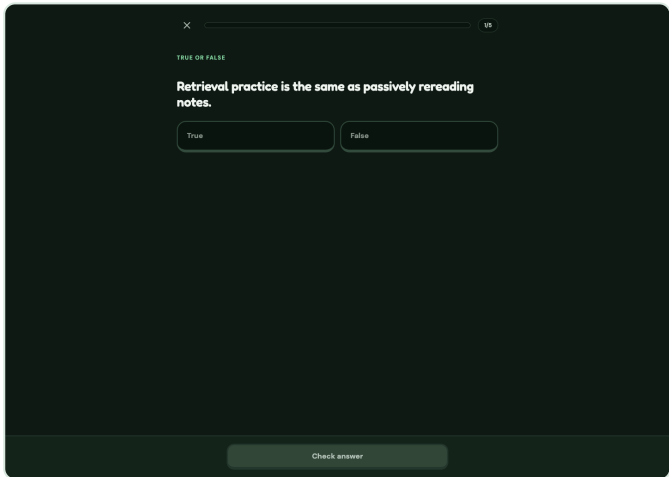


Figure 11 — Dark theme, quiz.

7.8 Responsive and tablet layouts

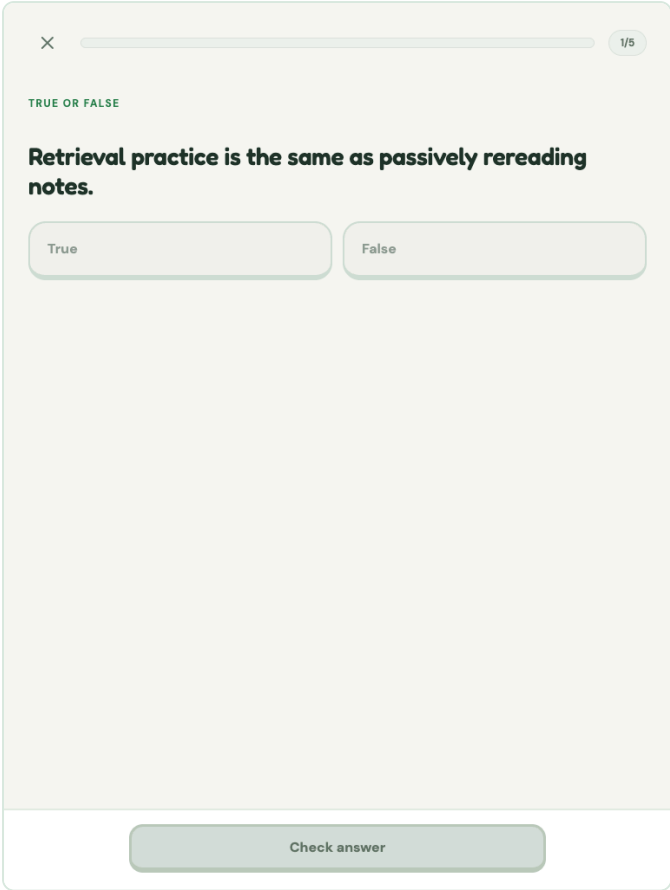


Figure 12 — Tablet layout, true/false item.

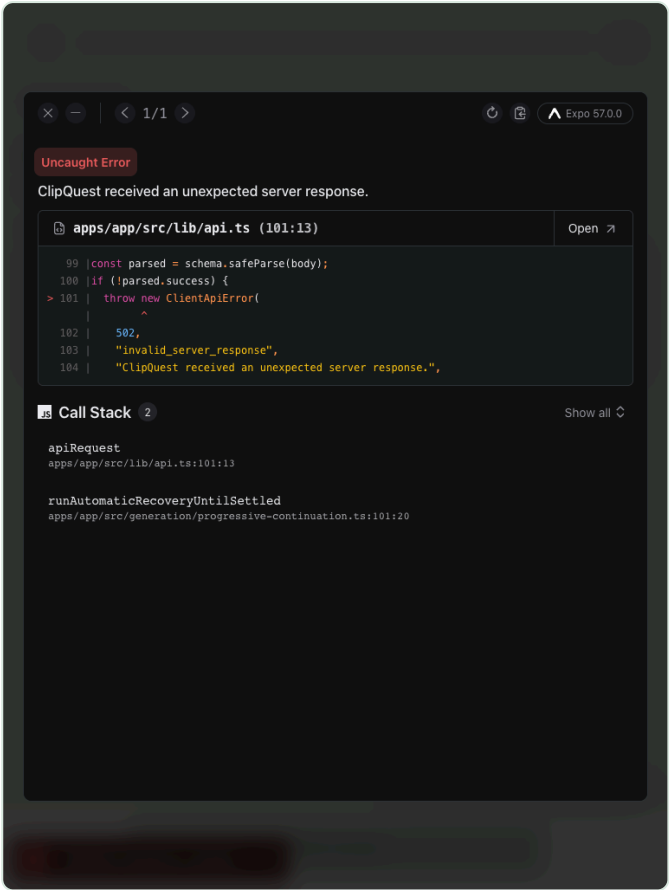
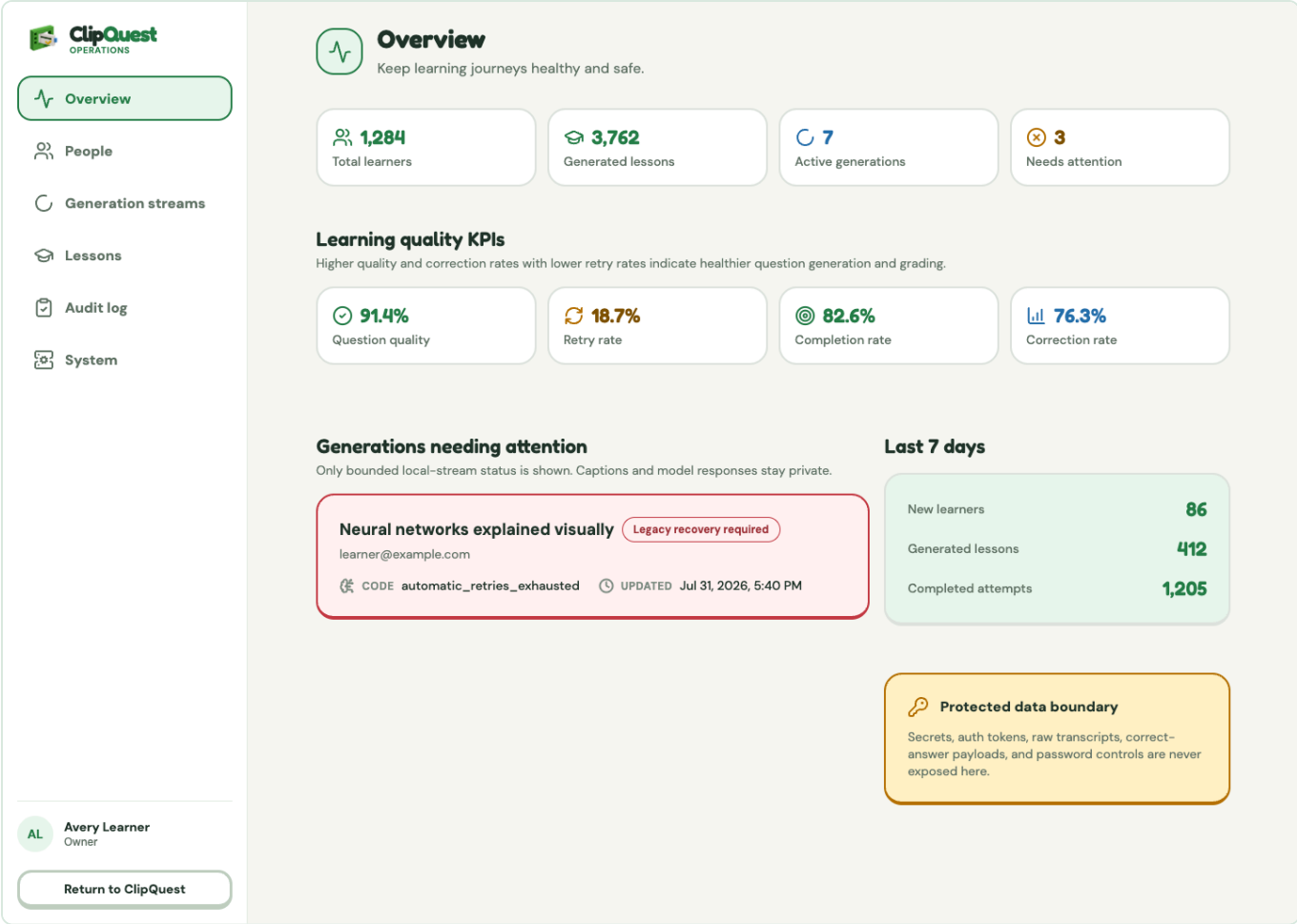


Figure 13 — Tablet layout, short-answer item.

7.9 Operations console

A private, role-gated console for processing overview and user administration, with audited privileged actions.



8. System architecture

ClipQuest is a monorepo with four runtime surfaces and two shared packages. The shared packages are what keep the surfaces honest: the contracts package is the single source of truth for every protocol, and the local quiz engine is shared verbatim between the Chrome extension and the native clients, so there is exactly one implementation of prompt construction, parsing, validation, and repair.

Layer	Technology	Responsibility
Web app	React 19, React DOM, TypeScript	Authentication, paste/import, lesson preview, quiz flow, feedback, completion, mastery, Library, leaderboard, profiles, administration, and themes
Browser extension	Chrome Manifest V3, JavaScript	Public caption access, local streamed DeepSeek generation, bounded repair, cancellation, and canonical option randomization
Edge API	Cloudflare Workers, Hono, scheduled triggers	Better Auth sessions, ordered storage-only imports, deterministic grading, availability, review scheduling, and static asset serving
Data and assets	D1 (SQL), KV, private R2	Accounts, videos, generating and passed banks, attempts, mastery, profile analytics, rate limits, thumbnails, avatars, and private PDFs
Local quiz engine	Shared JavaScript, DeepSeek V4 Flash	Progressive prompts, per-question validation, automatic repair, shuffling, and serialization
Shared contracts	TypeScript, Zod	Versioned page protocol, quiz schemas, API requests, and server-side validation
Quality and delivery	Vitest, Node test runner, Playwright, ESLint, Prettier, Wrangler	Contract tests, browser journeys, static checks, builds, and deployment
Visual identity	Semantic tokens, typed voxel registry, Sharp	Shared light/dark themes and deterministic web and extension assets

8.1 The generation protocol

The extension and the page communicate over a versioned result protocol with exact-origin boundaries, request IDs, payload bounds, cancellation, and timeouts. The current new-bank contract is result protocol `10`, capability `question-stream-v2`, pipeline `9`, prompt `quiz-local-json-stream-v5.2`, validator `validator-local-progressive-v4.1`, and progressive import `v4`. Every bank permanently records the prompt, validator, profile, model, and client-integrity metadata it was built with, so a bank generated last week is still interpretable after the pipeline moves on.

8.2 Why storage-only was the right server design

A conventional design would put generation on the server, where it is easy to control. We deliberately gave that up, and in exchange the Worker became small enough to reason about exhaustively: it authenticates, it validates, it stores in order, it grades deterministically. It has no Queue binding for generation and no fallback question path. This means the failure modes are enumerable — and a whole category of "the AI quietly made something up and the server saved it" bugs is structurally impossible rather than merely unlikely.

9. Technology stack

Technology	Where it is used
TypeScript	Primary language: web app, Cloudflare Worker, shared contracts, and the majority of tooling
React 19 + React DOM	The entire web product interface, routing, and state
JavaScript / ESM	Manifest V3 extension runtime, caption processing, local quiz generation, web workers, asset scripts, packaging
Cloudflare Workers + Hono	Edge API, routing, middleware, scheduled triggers, static asset delivery
Cloudflare D1 (SQL)	26 ordered migrations covering authentication, quiz storage, reliability, administration, progressive imports, safe call telemetry, recovery claims, and profile analytics
Cloudflare KV	Rate limits and short-lived operational state
Cloudflare R2 (private)	Thumbnails, avatars, and private PDF artifacts
Better Auth	Session management, accounts, and server-side role storage
Zod	Runtime schema validation for every protocol boundary and API request
DeepSeek V4 Flash	Question generation, called from the learner's browser with the learner's key, thinking disabled, temperature 0.2, JSON-object response mode
Chrome Manifest V3	Background service worker, tab coordination, page bridge, and scoped local key storage
Swift / Kotlin	Native iOS and Android client paths reusing the shared local quiz engine
Vitest + Node test runner	Unit, contract, API, app, engine, and extension tests
Playwright	End-to-end Chrome learner journeys
ESLint + Prettier	Repository-wide static analysis and formatting gates
Wrangler	Worker bundling, dry runs, migrations, and production deployment
Sharp	Deterministic generation of favicon, PWA, and extension artwork derivatives
HTML / CSS / Web App Manifest	Extension popup, QA harness, and installable PWA identity
GitHub Actions	Continuous integration across tests, types, lint, build, and packaging

10. Reliability, correctness, and recovery engineering

Moving generation into an untrusted browser created the central technical problem of this project: **how do you keep a 15-question quiz correct and ordered when any individual model call can fail, truncate, duplicate, or return the wrong type — and you refuse to ever fake a replacement?** This is where most of our engineering effort went.

10.1 Per-question independence

Each completed question object is parsed, validated, and imported on its own. Validation checks the expected ordinal, the requested type, the required fields, the answer mapping, and duplicate prompts. Because items are independent, **a malformed question 9 cannot erase questions 1–8**. The accepted prefix is preserved and remains answerable.

10.2 Bounded repair of the first missing ordinal

When a question fails validation, the engine asks DeepSeek to regenerate *only the first missing ordinal*, passing the exact validation reason and using bounded backoff. There is no unbounded retry loop, and there is no substitution of fallback content. If repair cannot succeed within its bounds, the failure becomes visible rather than being papered over. Accepted questions are never silently replaced.

10.3 Ordered, idempotent, ownership-checked imports

Every import requires ownership, a UUID idempotency key, a rate-limit allowance, strict pipeline metadata, and the exact next ordinal. Question 1 creates a generating bank and a full-length attempt immediately; later questions append while the learner is already answering. Skipped or conflicting ordinals fail closed. A bank becomes passed, reviewable, and Library-eligible only after all 5, 10, or 15 requested questions have been stored — so an incomplete bank can never be mistaken for a finished lesson.

10.4 Catching the generation frontier

Because the learner starts answering before the bank is complete, a fast learner can catch up to the generation frontier. When that happens the quiz waits and polls automatically with bounded backoff while the recovery coordinator continues the missing suffix. There is deliberately **no manual "continue" button** — an earlier iteration had one, and it turned an internal implementation detail into the learner's problem.

10.5 Anti-memorization randomization

Multiple-choice options are securely shuffled before import and shuffled again each time a learner view is activated. Display indexes are mapped back to canonical indexes before submission, so the correct answer is never inferable from position. True/False order stays True-then-False for predictability.

10.6 Deterministic grading

Short answers are graded by the authenticated Worker from a bounded stored rubric — one to three independent required ideas and three to six complete full-credit variants — with a 67% alternative threshold, plus normalization of pronouns, safe acronyms, and conservative domain aliases. Formula answers retain structural grading. Answer writes and mastery updates are guarded by an active grading reservation, and automatic generation appends require a live owner lease, so concurrent tabs cannot corrupt an attempt.

11. Privacy and security boundary

The privacy boundary is architectural, not a policy promise. Data that never crosses a wire cannot be leaked from the other side of it.

- The learner's DeepSeek key is stored only in `chrome.storage.local` and is sent only to `https://api.deepseek.com`. The ClipQuest page and Worker never receive it, and it is never part of any ClipQuest protocol.
- Caption segments, normalized text, prompts, and raw model bodies stay inside the local generation client. Import receives only video/session settings, verified aggregate source metadata, bounded generation metadata, and locally validated questions.
- The Worker performs no backend quiz generation and returns no generated-looking fallback questions. Invalid, incomplete, wrong-type, or malformed extension output fails closed.
- Call telemetry and recovery claims never store generation instructions, captions, raw model output, credentials, or DeepSeek errors.
- D1 queries and R2/KV objects are scoped to authenticated users. Operations roles are stored server-side, privileged changes are audited, and generic Better Auth admin endpoints are blocked.
- Short-answer grading is local to our Worker — the learner's prose is never forwarded to a model provider.
- Only public videos with usable text captions are supported. Missing captions are rejected before DeepSeek can run. ClipQuest never downloads, decodes, or transcribes audio.
- YouTube OAuth, watch history, subscriptions, playlists, liked videos, and personalized feeds are outside the core flow and disabled by default.

12. Original visual system and brand

ClipQuest's interface takes product-craft cues from polished learning software — focused tasks, visible progress, large controls, tactile depth, immediate feedback, generous whitespace — while using independently authored colours, geometry, components, and artwork. Our documented design research explicitly excludes any third-party assets, palettes, characters, proprietary fonts, sounds, copy, or traced illustrations.

System element	ClipQuest implementation
Brand anchor	Structural green <code>#247D49</code> with action green <code>#54C878</code>
Themes	Equally supported light and deep-green dark themes with semantic blue, warning, and error roles
Brand object	A non-anthropomorphic "learning prism" built from an interlocking video frame, quiz card, knowledge marker, and companion block — no face, body, limbs, pose, or mood
Primary lockup	The learning prism with a custom-cased Fredoka <code>ClipQuest</code> wordmark; constrained launcher icons keep the prism alone
Iconography	Individually generated, transparent, low-density isometric voxel PNGs resolved through a typed static registry — no stock glyph set
Typography	Fredoka for short display copy, DM Sans for body and interface copy
Interaction	44 px minimum targets, 16–24 px radii, visible keyboard focus, semantic feedback, and reduced-motion support

Platform identity

Deterministic derivatives for favicon, PWA, and browser-extension artwork

Processing, success, and error states are communicated by status copy, semantic panels, and progress — never by mascot behaviour. This keeps the interface legible to screen readers and to learners with reduced-motion preferences.

13. Iteration log — what changed during the 36 hours

The work was genuinely iterative rather than a single planned build. Each stage below addressed a concrete product risk that the previous stage exposed.

Iteration	Problem or goal	Improvement delivered
1. Product foundation	A written idea needed a usable learning flow.	Built the web app, authentication, URL import, lesson preview, quiz flow, feedback, completion state, and Library.
2. Evidence-grounded generation	AI questions needed reliable lesson context.	Added caption-only public YouTube acquisition, normalization, structured DeepSeek generation, strict schemas, and explicit caption failure states.
3. Progressive learning	Waiting for a complete quiz felt slow.	Added question-first progressive generation so the first validated question opens the attempt while the rest continue in bounded batches.
4. Correctness and recovery	A malformed question could break an attempt.	Added ordinal validation, duplicate checks, repair of the first missing question, idempotent imports, cancellation, timeouts, and safe recovery.
5. Mastery loop	A result alone did not create repeatable learning.	Added Library history, colour-coded mastery, review state, progress bars, due reviews, saved lessons, and practice presentation.
6. Cross-platform access	The experience needed to work beyond one browser screen.	Added a Chrome extension bridge plus native iOS and Android paths sharing the local quiz engine and the privacy boundary.
7. Product polish	The first UI needed stronger hierarchy and feedback.	Redesigned cards, hover and theme states, progress treatment, streaming feedback, empty and error states, responsive layouts, accessibility, and reduced motion.
8. Release readiness	A working feature needed repeatable delivery.	Added contracts, migrations, API routes, tests, asset verification, extension packaging, Wrangler deployment, health checks, and release documentation.

13.1 Improvement logic

- 1. Ground the content.** Use public captions rather than invented lesson content. Unsupported or captionless sources fail explicitly.
- 2. Protect correctness.** Validate every question before it becomes learner-visible or persistent. Accepted questions are never silently replaced.
- 3. Reduce waiting.** Progressive generation makes question 1 available quickly while preserving ordered completion.
- 4. Make progress meaningful.** Results become mastery, review, and Library state.
- 5. Make the product usable.** Responsive layouts, theme parity, accessible controls, visible states, and motion-safe feedback are part of the feature, not a later pass.
- 6. Ship with evidence.** Automated checks and production deployment are included; unfinished features remain clearly gated.

14. Pre-event versus post-event comparison

Area	Pre-event (one week before)	Post-event (submitted here)
Product state	Short README only	Deployed web product with a browser-extension bridge and native implementation paths
Input	Product idea	Public YouTube URL validation, metadata preview, and caption acquisition
AI	No implemented AI flow	Learner-owned DeepSeek key used locally for structured, schema-validated question generation
Learning	No quiz experience	Multiple-choice, true/false, and short-answer quizzes with immediate corrective feedback
Progress	No persistence or mastery model	Authenticated attempts, ordered storage, mastery ranks, review state, and Library
Reliability	No validation or recovery	Contract validation, bounded retries, first-missing-ordinal repair, cancellation, timeouts, and idempotency
UX	No interface	Responsive UI with light/dark themes, motion, accessibility, and clear states
Platforms	No working client	Web, Chrome extension bridge, iOS path, and Android path sharing core logic
Delivery	No deployment	Cloudflare Worker and static assets deployed at clipquest.ccwu.cc
Verification	No test suite	Automated contract, API, app, engine, extension, UI, build, and asset checks
Brand	No visual identity	Original palette, learning-prism brand object, custom lockup, voxel icon registry, and dual themes

15. Verification, QA, and honest release status

15.1 Automated source gate

The 22 August 2026 source gate passed **138 unit, contract, API, app, and extension tests** plus **21 Playwright Chrome journeys**, together with repository-wide TypeScript and ESLint checks, the web build, extension packaging, Worker bundling, and a Wrangler dry run.

15.2 Live production matrix — re-run

We re-ran the ten-video matrix on 22 August 2026 against **real public YouTube captions and the real DeepSeek API** using the headless QA harness (`npm run qa:quiz`) with a local key, ten questions per video, all three question types, and reference-answer grading enabled. Unlike the earlier run, this one actually exercised the prompt-first `v5.12` profile (`quiz-local-json-stream-v5.12` , validator `validator-minimal-gradeability-v5.3` , import `extension-progressive-import-v8` , protocol 10, pipeline 9).

The headline: **every video reached question 1, every stored question was unique, valid, and correctly graded, no invented content was ever emitted**, and the prompt-phrasing defect that blocked our previous release gate is **fully resolved**. The one metric still short of target is first-attempt full-bank completion, covered below.

Metric	Result
Videos attempted	10
Reached question 1 (progressive entry)	10 / 10
Complete 10-question banks, first attempt	5 / 10
Complete banks after one re-attempt	8 / 10
Questions stored across complete banks	80 (80 unique prompts, zero duplicates)
Model calls / automatic bounded retries	80 / 13
Fallback or invented questions emitted	0
Time to question 1 — median / mean / max	19.9 s / 22.0 s / 41.8 s
Deterministic grading on reference answers	80 / 80 correct
Validator, duplicate, fragmentary, and consistency audits	80 / 80 PASS each

Resolved: the content-acceptance regression

The defect that blocked our previous release gate is gone. Prompts containing "According to" fell from **79/100 to 0/80**, "According to the lesson" from **26/100 to 0/80**, and the display-time prefix sanitizer that visibly corrupted at least five prompts now corrupts **none**. All 80 stored prompts are unique and pass the production validator.

Known limit, fully diagnosed: full-bank completion

Five of the ten videos needed a second attempt to produce a complete ten-question bank. We have traced this to a **single, well-understood cause** rather than a diffuse quality problem: one `question_answer_kind_mismatch` outcome on one ordinal, repeated until its three bounded structural retries were spent.

- **The safety guarantee held in every case.** The engine failed closed: the accepted prefix was preserved, no fallback content was substituted, and no invented question was ever shown to a learner or written to storage. The system behaved exactly as designed — it would rather stop than teach something ungrounded.
- **The cause is isolated and reproducible.** Re-running the five recovered three, which tells us the behaviour is probabilistic rather than inherent to those lessons. Two videos reproduce it reliably, so we have a fixed test case to work against.
- **The fix is scoped.** Widening the retry budget and tightening the type-plan instruction for this one outcome is our next task; nothing about the architecture needs to change.

Net position: the re-run **clears** progressive entry, the storage-only privacy boundary, the no-fallback guarantee, content acceptance, and deterministic grading. The one gate still open is zero-intervention full-bank completion, and it is open for a single identified reason that we can point to in the logs.

We publish these numbers deliberately. A real, instrumented run against live captions and a live model is far stronger evidence than a green mock suite — our own Playwright journeys use contract-shaped mocked responses, and we say in our documentation that a green UI suite is *not* proof of live caption access or a completed learner flow. Every figure above is reproducible: the full artifacts, including per-call event streams for every run, are committed under `output/headless/matrix-2026-08-22/`, with the complete write-up in `docs/QA-HEADLESS-MATRIX-2026-08-22.md`. Judges can re-run the same harness with `npm run qa:quiz`.

15.3 Deliberately gated features

The in-development Workplace AI chat route is hidden from navigation for every user and redirects to Library. Its source stays behind a shared release gate. We chose to gate it rather than demo an unfinished feature.

15.4 Authoritative production checks

The live `/health` response and the Wrangler deployment history — not a version number typed into a document — are the authoritative production checks. Health exposes model and pipeline versions plus `backendQuizGeneration`, `extensionQuizGeneration`, and `extensionRequired` readiness flags without exposing secrets.

16. Team and responsibilities

Four members, one shared codebase. We worked with collective ownership of the core — every member committed to the web app, the API, and the extension — while each person led a primary area and owned the decisions in it. Commit counts below are from the public Git history.

Member	Primary role	Commits	Led
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UnoxyRich	Project lead & architecture	160	Product concept and scope, monorepo architecture, the caption-only privacy boundary, the edge API and D1 schema design, deployment and release management, and the production QA matrix
Layla	Frontend & design system	120	Web application interface, the learner journey, the original visual system, light/dark theme parity, responsive layouts, accessibility, motion, and the screenshot/asset pipeline
Justin-Yonardo	Generation engine & QA	111	The shared local quiz engine, prompt and validator design, bounded repair and recovery logic, Chrome extension internals, and the Playwright end-to-end journey suite
JimmyfaQwQ	Backend, data & platform	87	Worker routes, Better Auth integration, migrations, deterministic short-answer grading, mastery and review scheduling, the community/leaderboard layer, and the operations console

16.1 How we made decisions

- **Shared decisions:** core product concept and UX flow, feature scope and prioritization, project architecture and file structure, the caption-only privacy boundary, the key validation and recovery algorithms, the mastery and review model, testing strategy and CI, and the visual and accessibility direction.
- **Conflict resolution:** when the progressive-generation design collided with ordered storage, we resolved it by writing the contract first and letting the contract arbitrate — the versioned protocol in `packages/contracts` is the artifact of that decision.
- **Learning:** the manual "continue" button was our clearest reversal. We shipped it, watched it expose an internal implementation detail to learners, and removed it in favour of automatic bounded polling.
- **Defence coordination:** each member presents the area they led, with the project lead handling architecture questions and the engine lead handling correctness and failure-mode questions.

17. AI usage disclosure

Declared per the HHCC 2026 Participant Handbook's AI Usage Guidelines.

Item	Declaration
Tools used	GitHub Copilot and Claude Code CLI
Usage cost	Approximately \$240 in API-token usage across all tools
Purpose	Boilerplate and repetitive code, syntax lookup and autocomplete, refactoring and code formatting, documentation drafting, and debugging assistance
Team-originated	Core product concept and UX flow; feature scope and design decisions; project architecture and file structure; the caption-only privacy boundary; key algorithms including validation, repair, ordering, and grading; the mastery and review model; testing strategy and CI setup; and all visual and creative direction
Not done by AI	No module was submitted that the team does not understand. Every member can explain the principles and implementation of the code in their area on request during Q&A.

All implementation code for the event iteration was written in this project repository during the official hacking period. AI served strictly as an auxiliary coding tool; all key decisions, concepts, and creative direction were developed by the team.

18. Social impact, sustainability, and business model

18.1 Who this is for

Self-directed learners who already rely on free educational video: secondary and university students revising from recorded lectures, self-taught developers and designers, exam candidates, and adult learners in regions where paid course platforms are out of reach. ClipQuest requires no institution, no enrolment, and no curriculum licence — only a link.

18.2 Why the economics work

Because inference runs in the learner's browser on the learner's own key, our marginal cost per generated quiz is **zero**. The server does authentication, validated storage, and deterministic grading — all of which run comfortably on Cloudflare's edge at a cost that scales with storage rather than with tokens. This is what allows the core learning loop to stay free without a paywall on the thing that matters.

Dimension	Position
Core loop	Free, learner-key powered, no per-quiz charge
Future revenue	Optional managed-key convenience tier, classroom/cohort dashboards for educators, and institutional deployment of the same self-hosted stack
Content sustainability	Uses public captions of videos the learner already has access to; creates no scraped content library and stores no transcripts

Environmental cost	Edge-only backend, no always-on servers, no audio download or transcription, and small non-thinking model calls at temperature 0.2
Risk handling	Public-caption-only sourcing, no OAuth into personal YouTube data, fail-closed on missing captions, and audited privileged operations
Openness	Apache-2.0 licensed and fully public, so institutions can self-host and audit it

18.3 Accessibility as social value

Accessibility was treated as part of the feature rather than a compliance pass: 44 px minimum targets, visible keyboard focus, semantic (not colour-only) feedback states, reduced-motion support, full light/dark parity, and responsive layouts down to tablet and mobile. A learning tool that only works for able-bodied users on a large desktop screen is not an accessible learning tool.

19. Current progress and future plans

19.1 Shipped and live today

- Deployed, publicly reachable product at <https://clipquest.ccwu.cc>.
- End-to-end learner flow: import → caption acquisition → progressive generation → quiz → feedback → recap → Library.
- Three question formats with deterministic short-answer grading.
- Mastery, review scheduling, saved lessons, leaderboard, public profiles, and activity calendar.
- Chrome extension bridge, installable PWA, and native iOS/Android paths on the shared engine.
- Role-gated operations console with audited privileged actions.
- CI across tests, types, lint, build, packaging, and a Wrangler dry run.

19.2 Immediate next steps

1. **Close out `question_answer_kind_mismatch`**. This single outcome is the last thing between us and a clean matrix, and it is already isolated: widen the retry budget and tighten the type-plan instruction for this one case. AP Physics 1 Kinematics (`qP-9wwRrJbg`) and AP Calculus AB Unit 1 (`G8zkXA5TXgg`) reproduce it reliably and serve as the regression tests.
2. Raise first-attempt full-bank completion from 5/10 to 10/10, then re-run the matrix to confirm the zero-intervention gate is met.
3. Promote the now-validated prompt-first v5.12 profile through the staged rollout — an immutable extension candidate, a canary assignment, and a fresh benchmark — since the content-acceptance regression that previously blocked it is resolved.
4. Investigate the 4 `absolute_wording` NOTICE results, which are advisory today but indicate prompt phrasing worth tightening.
5. Finish captionless local transcription on supported hardware, and Resend email delivery.

19.3 Longer-term direction

- Educator mode: assign a video to a cohort and see aggregate concept-level mastery.
- Cross-lesson knowledge graph so review scheduling works across related videos, not per video.
- Additional model providers behind the same local-key boundary, so learners are not locked in.

- Firefox and Safari extension parity.
- Re-enabling the gated Workplace AI chat once it meets the same evidence-grounding bar.

20. Appendix — repository structure and links

```
ClipQuest/ └─ apps/ | └─ app/ # Web client, routes, components, themes,
static export | └─ extension/ # Manifest V3 captions and local-AI
extension | └─ api/ # Cloudflare Worker, bindings, and 26 D1 migrations └─
packages/ | └─ contracts/ # Shared Zod API and quiz schemas (single source
of truth) | └─ local-quiz-engine/ # Prompt, parser, validation, and retry
core └─ e2e/ # Playwright learner journeys └─ docs/ # Product guides, QA
reports, design research, screenshots └─ output/ # Generated QA artifacts
and reports
```

Resource	Location
Live product	https://clipquest.ccwu.cc
Repository	https://github.com/404-Waifu-Not-Found/HHCC2026
Engineering reference	<code>README.md</code> — includes a judge briefing mapped to the scoring dimensions
Iteration overview	<code>docs/ITERATION-OVERVIEW.md</code>
Learning-science brief	<code>docs/HACKATHON.md</code>
Headless matrix re-run	<code>docs/QA-HEADLESS-MATRIX-2026-08-22.md</code> — the section 15.2 evidence
Live QA report	<code>docs/QA-YOUTUBE-BROWSER-10X-2026-08-22.md</code>
Progressive generation QA	<code>docs/QA-FIRST-QUESTION-ETA-2026-08-22.md</code>
Production release guide	<code>docs/PRODUCTION-RELEASE.md</code>
Design research boundary	<code>docs/duolingo-ui-research.md</code>
Operations console	<code>docs/ADMIN-CONSOLE.md</code>
Licence	Apache-2.0

Verifying our in-event claim

To confirm that all product code was written during the 36-hour period, clone the repository and run `git log --format='%ad %an %s' --date=iso --reverse`. The first product commit and the last both fall inside 21–22 August 2026. Nothing was squashed, back-dated, or imported from a prior repository.